

EXPO MTPM Replacement — Requirements Document

2026-06-23

APPROVED by Richard Patch 6/23/2026

Points of Contact

Role	Name
Customer / Primary User Contact	Richard Patch
RAG Calculation Owner	Griffin Buscavage
Developers	Ryan Clark, Davis Blanch

Tool History

Period	System	Notes
Pre-2008	Motorola/GDDS/GDC4S Project Workbook ("Level 5 Tool")	TPM and WPM metrics tracked in "Parameter" worksheets.
~2008	Enterprise Project Management System (EPMS) + EXPO	Web-based tool; TPM/WPM creation, updates, and project-to-org rollup.
2008–2020	EPMS / EXPO	12-year operational run.
2020	EPMS / EXPO retired	Adobe Flash end-of-life forced retirement.
2020–present	SharePoint + Power BI ("EXPO")	Temporary replacement — the system being replaced by this effort.

Key Implication: The current SharePoint/Power BI tool was a temporary migration of convenience — not a purposeful design. The new system shall be designed from scratch. The SharePoint list structure is not a useful architectural reference.

Executive Summary

The EXPO MTPM tool is a legacy SharePoint list used by program management stakeholders to track program performance metrics across 78 active projects and approximately 140 users. Each record captures actual, baseline, and forecast values with RAG (Red/Yellow/Green) threshold status.

This effort replaces the SharePoint list with a **database-backed application with a dashboard frontend**, hosted within the existing GDMS M365 GovCloud environment.

This tool was selected for rewriting because the effort is minimal — low cost, low risk, ideal for testing modernization processes.

System Context

Item	Detail
Active Projects	78
Total Users	~140 (136 Members + 4 Owners)
Primary User Roles	DPMEs and Lead Engineers
LOB Coverage	Primarily Land & Air Systems; all three LOBs represented
Recording Cadence	Monthly, quarterly, semi-annually, or event-driven
Records Entered in 2025	315

Estimated Cost Savings

Updated by Dan Harlan following discussion with Richard Patch (June 2026). Previous estimate of ~\$2,900 was based on current user base only and did not account for growth. Revised estimate reflects projected growth of 150 projects/year at ~3 users per project. Current system could not handle this projected capacity.

Category	Basis	Estimate
Maintenance savings	None assumed — new application; costs will grow with usage	\$0
Performance savings	Avg. 365 users × 15 min/user × \$90/hr (unburdened engineering rate)	\$8,212
Data entry savings	Avg. 450 projects × 10 records/month × 15 min/record × \$90/hr	\$101,250
Total estimated annual savings		\$109,462

Power BI Connectivity

Power BI connectivity is **in scope** for this effort. The existing Power BI dashboard must remain functional after migration. The dev team is responsible for ensuring the AKS-hosted replacement database supports an external Power BI connection.

Confirmed at the June 1 kickoff by Dan Harlan ("Power BI reporting should stay as is, will need to connect to new source", AISAVING-137) and separately via email from Richard Patch.

Dev team responsibilities:

- Design the database schema to expose data in a format compatible with the existing Power BI dashboard
- Configure the AKS-hosted database to support an external Power BI connection
- Validate that the Power BI dashboard successfully connects to the new data source prior to go-live

Data Model

Projects

Field	Type	Description
Project ID	Integer (PK)	System-generated unique identifier.
GUPI	Integer (unsigned)	GDMS Unique Project Identifier — links to the GDMS Project Registry (workflow.gd-ms.com). Primary external reference. Per Richard Patch: Project Name should map to the GUPI from the Project Registry. [Type confirmed by Richard Patch, June 22, 2026]
GUPI(s)	Integer (unsigned, multi)	Additional GUPIs for follow-on projects carrying over TPMs/WPMs (e.g., CPM Plus TO 1 → TO 30). Supports multiple Project Registry entries per EXPO project. [Type confirmed by Richard Patch, June 22, 2026]
Project Name	String	Display name — sourced from or matched to the GDMS Project Registry entry.
Line of Business	String	LOB this project belongs to. Also in Project Registry — may be pulled automatically once GUPI is selected.
Is Private	Boolean	Restricts visibility to project members. Default: false (public). IT involvement required to define visibility rules — certain roles (e.g., ELT, IT Admins) may need permanent visibility regardless of private setting. Per Richard Patch, June 9, 2026.
Created By	User ref	User who created the project record.
Created On	DateTime	Timestamp of creation.

Measurements Reference

Measurement definitions are **unique per project** — not globally shared.

Field	Type	Description
Measurement ID	Integer (PK)	System-generated unique identifier.
Project ID	FK → Projects	The project this measurement belongs to.
Measurement Name	String	Display name of the measurement.
Units	String	Unit of measure (e.g., %, count, days).
Direction	String	Valid values: Upward, Downward, N/A. Applies to WPMs only. [Q-6: Resolved]

Green Threshold	Decimal	Threshold value for Green status.
Yellow Threshold	Decimal	Threshold value for Yellow status (outside = Red).
Dropdown	String	[Q-8: Open — Richard not sure. Dev team to show EXPO example at meeting.]
Is Active	Boolean	Whether this measurement is available for new records.

Measurement Records

Field	Type	Description
Record ID	Integer (PK)	System-generated unique identifier.
Project ID	FK → Projects	The project this record belongs to.
Measurement ID	FK → Measurements	The measurement type for this record.
Title	String	Name/label for this specific measurement entry.
Measurement Date	Date	The date the measurement was recorded.
Description	String	Narrative description of the measurement entry.
Actual	Decimal	The actual measured value.
Baseline	Decimal	The baseline/target value.
Forecast	Decimal	The forecasted value.
AvB	Decimal	Actual vs. Baseline. Calculated as Actual – Baseline. [Q-2: Resolved — calculated only.]
Items	Integer	[Q-7: Open — Richard not sure. Dev team to show EXPO example.]
Percent Complete	Decimal	[Q-3: Resolved — calculated, not manually entered. Formula TBD with Griffin Buscavage.]
RAG Status	Enum (Red/Yellow/Green)	Derived from Actual, thresholds, and Direction. WPMs: threshold logic. TPMs: may use CPK value. [Q-4: Required before prototype.]
Show Actual	Boolean	Controls whether Actual is displayed in dashboard views.
Show Forecast	Boolean	Controls whether Forecast is displayed in dashboard views.
Show Percent Complete	Boolean	Controls whether % Complete is displayed in dashboard views.
Version URL	String (URL)	[Q-9: Open — Richard not sure where this appears in EXPO.]
Created By	User ref	User who created the record.
Created On	DateTime	Timestamp of creation.
Modified By	User ref	User who last modified the record.
Modified On	DateTime	Timestamp of last modification.

Percent Complete History

Field	Type	Description
History ID	Integer (PK)	System-generated unique

		identifier.
Record ID	FK → Measurement Records	The measurement record this history belongs to.
Percent Complete	Decimal	The % complete value at time of recording.
Recorded On	DateTime	Timestamp when this value was captured.
Recorded By	User ref	User who recorded this value.

Legacy Audit Fields (Migration Only — Read-Only)

Field	Type	Description
Legacy Item ID	String	Identifier from the prior system.
Legacy Created By	String	Created-by value from prior system.
Legacy Created On	DateTime	Creation date from prior system.
Legacy Changed By	String	Last-changed-by value from prior system.
Legacy Changed On	DateTime	Last-changed date from prior system.

Functional Requirements

1. Measurement Record Management

1.1 Users shall be able to create, read, update, and delete measurement records, subject to their role.

1.2 When creating a project, users shall search and select from the GDMS Project Registry by GUI. Project Name and Line of Business shall be sourced from the Registry automatically upon GUI selection. Users shall be able to associate multiple GUIs with a single EXPO project to support follow-on project scenarios (e.g., CPM Plus TO 1 and CPM Plus TO 30 carrying over the same TPMs/WPMs). Per Richard Patch, June 9, 2026.

1.3 Required fields enforced on save: Measurement Name, Project, Measurement Date, Forecast. Baseline and Actual are NOT required — users may set a baseline for a future period before an actual value is measured. [Q-1: Partially resolved]

1.4 System prevents save if required fields are empty and displays a clear validation message.

1.5 Data entry shall be fast and responsive. Records loaded on demand or paginated — not all historical data upfront.

1.6 New record pre-populates from the user's most recent record for this measurement.

1.7 Static fields (thresholds, spec limits) carry forward from the previous record for the same project and measurement. User can override any pre-populated value.

1.8 New record entry is a single screen or inline form, similar to spreadsheet entries.

1.9 Date selection auto-saves on pick — no separate save action required after calendar selection.

1.10 Records presented in an editable table view. Inline or modal editing preferred over full-page navigation.

1.11 Users can set a default project pre-selected on login.

1.12 First-time users guided by inline hints or tooltips — no external training required.

2. Measurements Reference List

2.1 Measurements table maintained per project — scoped to the project, not globally shared.

2.2 Each measurement stores: Name, Units, Direction, Green Threshold, Yellow Threshold.

2.3 Only Owner or Admin roles can add, edit, or deactivate measurement definitions. [Q-11: Resolved — Owners only initially; Member delete deferred to future version.]

3. RAG Threshold Logic

3.1 RAG status auto-calculated from Actual, Green/Yellow thresholds, and Direction. WPMs use threshold logic. TPMs may calculate RAG from CPK value — confirm with Griffin Buscavage. Direction applies to WPMs only. (Confirmed by Richard Patch, June 22, 2026.)

3.2 Full CPK evaluation logic to be obtained from Griffin Buscavage before implementation. [Q-4 — required before prototype]

3.3 Red status implied when Actual falls outside the Yellow threshold. [Q-5: Resolved — Yes.]

3.5 RAG recalculates automatically when Actual, thresholds, or Direction values change.

4. Calculated & Derived Fields

4.1 AvB calculated automatically as Actual – Baseline. [Q-2: Resolved — calculated only, not user-settable.]

4.2 % Complete is calculated, not manually entered. [Q-3: Resolved — formula TBD with Griffin Buscavage.]

4.3 % Complete history stored as a structured table — not a delimited string.

5. Dashboard & Views

5.1 Dashboard supports filtering by: project, date range, RAG status, LOB, and measurement type.

5.2 RAG status indicators (color-coded) displayed prominently for each record.

5.3 Per-record display flags (Show Actual, Show Forecast, Show % Complete) control column visibility.

5.4 The new dashboard shall replicate the most commonly used data layouts and filter configurations from the current EXPO SharePoint tool. Richard Patch to identify which current views or saved filter configurations are highest priority to recreate. [Q-10: Open — dev team to show Richard examples of current EXPO layouts at next meeting to confirm priorities.]

5.5 Dashboard remembers each user's last-used filter selections across sessions.

5.6 Per-project history view shows all records over time and allows entry of a new value from the same screen.

5.7 Export of all records for a project to Excel or CSV. [Q-21: Confirm whether bulk import is required for v1]

5.8 Inactive projects are archivable; hidden from default views. [Q-22: Resolved — Owners can archive; may auto-archive at Project Registry end date + 60 days.]

5.9 Charts and visualizations render promptly — chart loading shall not block page interactivity.

5.10 The replacement database shall expose data compatible with the existing Power BI dashboard. Dev team configures AKS-hosted database for external Power BI connection.

6. Access Control & Permissions

6.1 Authentication via GDMS M365 / Azure AD — no separate login required.

6.2 AD group / SailPoint integration deferred to future release. [Q-13: Resolved — group permissions needed; open to alternatives.]

6.3 Roles:

Role	Permissions
Admin	Full access: manage all projects, users, measurements, and records.
Owner	Add Members to owned projects. Manage Measurements reference list. Full CRUD on owned project records.
Member	Create and edit records for assigned projects. Delete deferred to future version. [Q-11: Resolved]
Viewer	Read-only access to all public projects. Default for all authenticated users.

6.4 Projects default to public. Owners may mark private — restricts visibility to project members only. IT involvement required to define what "private" means for elevated roles — certain users (e.g., ELT, IT Admins) may require permanent visibility to all projects regardless of private setting. Per Richard Patch, June 9, 2026.

6.5 LOB-level group access deferred to future release. [Q-12: Resolved — deferred for V1.]

6.6 All create, update, and delete actions logged: user, action, timestamp, record ID.

7. Data Migration

7.1 Full migration required — all existing data including full history. Clean start is NOT acceptable. (Confirmed: Richard Patch)

7.2 Dev team exports SharePoint list to CSV/Excel. Davis Blanch provides Richard Patch with full legacy field inventory.

7.3 All records preserve: Measurement Date, Created On/By, Modified On/By, and all Legacy audit field values.

7.4 Legacy audit fields preserved as read-only in the replacement system.

7.5 Migration validation report generated post-import comparing source vs. destination record count and key fields.

7.6 Migration must be completed and validated by a stakeholder before go-live.

Transition Plan

Richard Patch specifically requested a formal transition plan prior to go-live. The following items are required and will be developed in detail during the prototype and UAT phases.

Item	Description	Status
User Notification	All current EXPO users must be notified of the new system prior to go-live, including instructions on how and when to begin using it. Richard Patch has requested to be included in crafting this message — coordinate with Richard before finalizing any notification. Per Richard Patch, June 22, 2026.	To be planned — Richard Patch to co-author
Data Freeze Period	A freeze period must be established during which users do not enter data into the legacy SharePoint tool while data is being migrated. Duration and timing TBD.	To be planned
Training	Training documentation and/or sessions must be provided to users prior to or concurrent with deployment.	To be planned — AC 6
UAT Notice	Richard Patch requires several weeks advance notice before UAT begins to identify and coordinate testers from his Business Areas.	Required — notify Richard at least 3 weeks before UAT
Awareness Timeline	A timeline for notification and awareness to the user community must be created and shared with Richard Patch before go-live.	To be planned
Legacy Decommission	Legacy EXPO SharePoint tool to be placed in read-only mode	Per AC 7

	following confirmed user adoption, then fully decommissioned.	
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Non-Functional Requirements

NFR-1: Performance

- Main project list loads within 3 seconds.
- Record create/save completes within 2 seconds.
- Data loaded on demand / paginated — no all-history fetch on load.
- Charts load asynchronously — do not block page interactivity.
- Responsive under concurrent use by up to 250 users. (Current: ~140. Enterprise target: ~10,000 total, 250 concurrent.)

NFR-2: Availability

- Target availability consistent with GDMS M365 GovCloud tools.
- No data loss on save failure — user notified, input preserved for retry.

NFR-3: Security

- Authentication via GDMS M365 / Azure AD — no separate login.
- Role-based access control enforced throughout.
- All data transmitted over HTTPS.
- Hosted within GDMS M365 GovCloud tenant — no data leaves boundary.
- All create/update/delete actions logged with user, timestamp, action, and record ID.

NFR-4: Usability

- Usable in GDMS-approved browsers without plugins.
- First-time users self-service via contextual guidance.
- User preferences (default project, last filter) persist across sessions.

NFR-5: Maintainability

- Source in version-controlled repository.
- Tech stack uses GDMS-approved, supported technologies — selection documented and approved before build begins.
- Deployment follows a documented, repeatable process.

NFR-6: Data & Reporting

- Database schema designed for direct Power BI connectivity.
- Export to Excel/CSV supported.
- Migrated records retain original timestamps and authorship.

Business Rules

Rule	Description
Required Fields	At minimum: Measurement Name, Project, Measurement Date, Forecast. Baseline and Actual are NOT required — users may set a baseline for a future period before an actual value exists. Full confirmed list TBD with Richard Patch.
RAG Evaluation	Calculated from Actual, thresholds, and Direction. Direction applies to WPMs only — confirmed by Richard Patch, June 22, 2026. Full RAG/CPK logic owned by Griffin Buscavage — obtain before implementation.
Measurements Are Project-Scoped	Measurement definitions unique per project, not shared globally.
Project Visibility	Default public. Owners may mark private.
Full Migration Required	All existing data including history migrated. Clean start NOT acceptable.
Requirements Gate	Prototype development shall NOT begin until requirements approved by Richard Patch.
Power BI Connectivity	Database must expose data compatible with the existing Power BI dashboard. Dev team configures external Power BI connection on AKS. In scope for this effort.

Access Control — Confirmed Rules (from Dan Harlan / kickoff meeting)

- Owners add Members to their projects.
- Members can add and edit records for their projects.
- Delete permission for Members — pending Griffin Buscavage confirmation.
- All authenticated users have view access to all public projects by default.
- Projects start as public but can be marked private at the project level.
- AD group / SailPoint integration is a future desire, not v1.
- LOB-level group access is a future option, not committed for v1.

Project Delivery States

State	Description
Discovery	Requirements gathering in progress. Kickoff with Richard Patch: 2026-06-01.

Requirements Approved	Richard Patch sign-off required. Prototype SHALL NOT begin before this gate.
Prototype	Core workflows built and validated. Migration completed and verified before UAT.
UAT	UAT with representative users. All critical/high defects resolved before deploy.
Deployed	Live in production. All EXPO functionality available.
Stabilizing	Post-go-live; legacy not yet decommissioned.
Closed	Legacy decommissioned. AISAVING updated. DoD signed off.

Acceptance Criteria

AC 1 Requirements formally documented, reviewed, and approved by Richard Patch.

AC 2 Working prototype demonstrates core EXPO workflows, validated against all Functional Requirements.

AC 2.1: Prototype validated against all Functional Requirements in this document.

AC 3 All existing EXPO MTPM data successfully migrated and verified prior to go-live.

AC 3.1: Davis Blanch provides Richard Patch with full legacy field inventory.

AC 3.2: Migration validation report produced and reviewed by Richard Patch.

AC 4 UAT conducted with a representative user sample. Richard Patch requires several weeks notice to identify testers.

AC 4.1: All critical and high-severity defects resolved before go-live.

AC 5 Replacement application deployed to production — all user-facing EXPO functionality available.

AC 6 Training documentation and/or sessions provided prior to or concurrent with deployment. Transition plan and user notification timeline required.

AC 7 Legacy EXPO SharePoint tool decommissioned or placed in read-only mode following confirmed user adoption.

Definition of Done

- ☐ All seven acceptance criteria met and signed off by Richard Patch and the product owner.
- ☐ All child stories in Done status with no open critical or high defects.
- ☐ Migrated data validated by stakeholder — confirmed consistent with source.
- ☐ Deployment stable in production for the agreed stabilization period — no rollback required.
- ☐ User base notified of transition; training materials accessible in a shared location.
- ☐ AISAVING feature updated with links to requirements, UAT results, migration assessment, and deployment notes.
- ☐ Power BI dashboard successfully connects to the new AKS-hosted database and validated prior to go-live.

Out of Scope — v1

- Modification of existing Power BI report visuals or report-level logic — only the data source connection is in scope.
- Net-new features not present in existing EXPO tool.
- Infrastructure procurement outside existing GDMS M365 GovCloud tenant.
- Full SailPoint / AD group integration (future direction).
- Search/filter at top of lists (assess during prototype).
- Digital PM Dashboard integration (future desire).

Open Issues — Action Required

Questions for Richard Patch

#	Question	Status
Q-1	Full list of required vs. optional fields for a measurement record.	Partially resolved — Baseline and Actual NOT required. Forecast IS required. Full list TBD.
Q-2	Is AvB always Actual – Baseline, or can it be manually entered?	Resolved — Calculated only; not user-settable.
Q-3	Is % Complete calculated automatically or entered manually?	Resolved — Calculated. Formula TBD with Griffin Buscavage.
Q-5	Is Red status implied when Actual falls outside Yellow threshold?	Resolved — Yes.
Q-6	Valid values for Direction?	Resolved — Upward, Downward, N/A. WPMs only.
Q-7	What is the Items field used for?	To be confirmed during prototype discovery — neither Richard nor the dev team could confirm the field purpose from the current system.
Q-8	What is the Dropdown field used for?	To be confirmed during prototype discovery — neither Richard nor the dev team could confirm the field purpose from the current system.
Q-9	What does versionURL link to? Is it still used?	To be confirmed during prototype discovery — neither Richard nor the dev team could locate this field in the current system.
Q-10	Which data layouts and saved filter configurations from the current EXPO SharePoint tool are highest priority to replicate? ("Legacy views" = preset filter/column configurations in the current SharePoint list.)	Open — dev team to show Richard examples of current EXPO layouts at next meeting to confirm priorities.
Q-11	Can Members delete their own records?	Resolved — Owners only initially. Member delete deferred to future version.
Q-12	LOB-level group access for v1?	Resolved — Deferred for V1.
Q-13	AD group / SailPoint integration for v1?	Resolved — Group permissions needed; open to alternatives.

		Team to recommend.
Q-14	Agreed stabilization period before legacy SharePoint decommission?	Open
Q-15	Does current tool send email alerts?	Resolved — No, never did. Deferred for V1.
Q-20	Provide improvements list and historical PowerPoint.	Resolved / Closed
Q-21	Is bulk import from Excel/CSV required for v1?	Open
Q-22	Who can archive a project?	Resolved — Owners. May auto-archive at Project Registry end date + 60 days.
Q-PBI	Power BI connectivity in scope?	Resolved — Yes. Dev team configures external connection on AKS.

Questions for Griffin Buscavage

#	Question	Status
Q-4	Full RAG/CPK evaluation logic — how is Green/Yellow/Red derived?	Open — required before prototype
Q-11	Can Members delete their own records?	Open

Internal Dev Team

#	Item	Owner	Status
#16	Davis Blanch provides Richard Patch with full legacy field inventory.	Davis Blanch	Open
#17	Reach out re: workflow request #207100. Mike Monteilh retired 2024. Contact Richard Patch.	Blanch / Clark	Open — contact Richard Patch
#18	Determine whether AISAVING feature link updates are dev team responsibility.	TBD	Open
#19	Tech stack selection — document with rationale, obtain product owner approval before build.	Blanch / Clark	Open (AISAVING-130)

Assumptions

- Hosted in existing GDMS M365 GovCloud — no new infrastructure required.
- Azure AD is the identity provider.
- Richard Patch available for review of this document before prototype begins.
- Existing EXPO SharePoint list is exportable to CSV/Excel.
- Dev team has or will have read access to existing EXPO tool.
- Griffin Buscavage will provide CPK RAG logic before prototype implementation.

- SharePoint list architecture is not a useful design reference — new data model designed from scratch.

User Feedback — Top Themes (19 Items Collected)

Theme	Verbatim	Requirement
Performance	"The site is generally slow — probably loading too much data. Charts take a long time."	1.5, 5.9, NFR-1
Data Carry-Forward	"Remember previous month's data. The measurement value is usually the only thing that changes."	1.7
Filtering / Default	"Filtering by project would help. Even better if it remembered my default filter."	5.5, 1.11
Entry Flow	"Adding a measurement is cumbersome — multiple screens, re-selecting project and metric every time."	1.6, 1.8
Inline Editing	"If data is entered incorrectly, it is cumbersome to find and update."	1.10
History + Entry	"See the history for a TPM and enter the next value from the same screen."	5.6
Export & Import	"Export TPMs for a project, add new values, and import it back."	5.7
Archiving	"Archive old projects no longer in use."	5.8

Authors: Ryan Clark, Davis Blanch